

ECE284 FA25 Final Progress Report

1. Fill the table below. This helps us to understand your current progress status.

Item	Current Status	Status during Poster Presentation	Note
Part1	Complete	Complete	
Part2	Complete	Complete	Using debug data
Part3	Complete	Complete	
Alpha 1 (Hybrid Pruning)	Complete	Complete	Perform unstructured and structured pruning together to reach a better accuracy and preserve regularity
Alpha 2 (Approx. Multiplier for Low Power)	Complete	Complete	Replace multiplication with approximate version to reduce power of MAC while inducing negligible error.
Alpha 3 (Region-aware Activation Extraction)	Complete	Complete	Selecting necessary features for each Kij in the kernel to reduce unnecessary psum computation.
Alpha 4 (Hybrid Read Mode and Pipelined RD/WR)	Complete	Complete	Allowing FIFO to select different read signal propagation for each row of FIFO and also supporting pipelined Read/Write operation to minimize data communication

			overhead.
Alpha 5 (Zero Skipping MAC Operation)	Complete	Complete	Gating the FF to stop data changes when weight or input are zero.